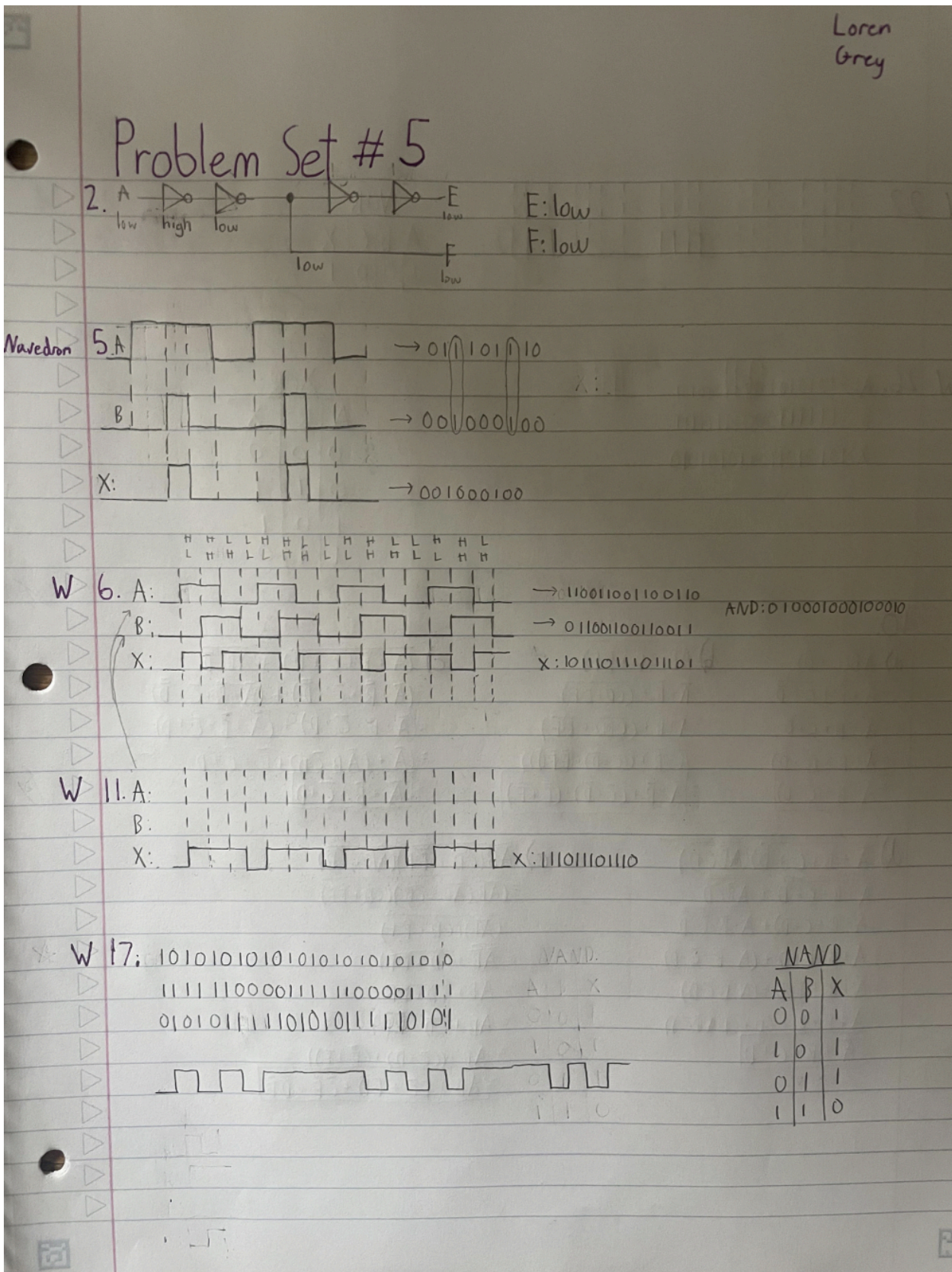
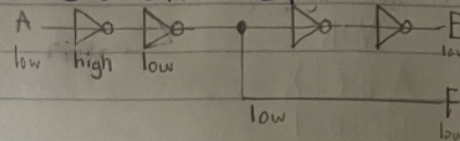


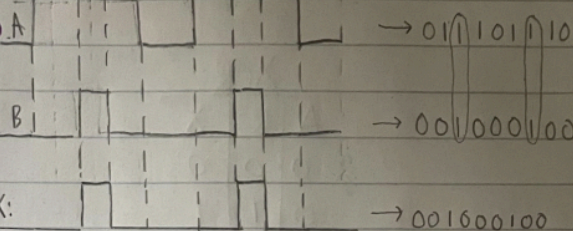
[illegible][illegible][illegible][illegible][illegible]

Problem Set #5

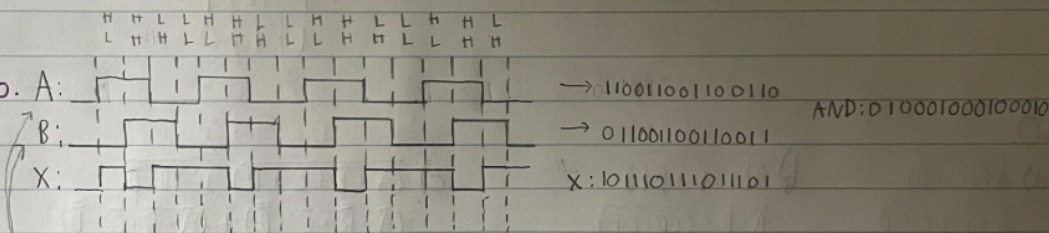
Loren
Grey

2. 

E: low
F: low

Navedron 5. 

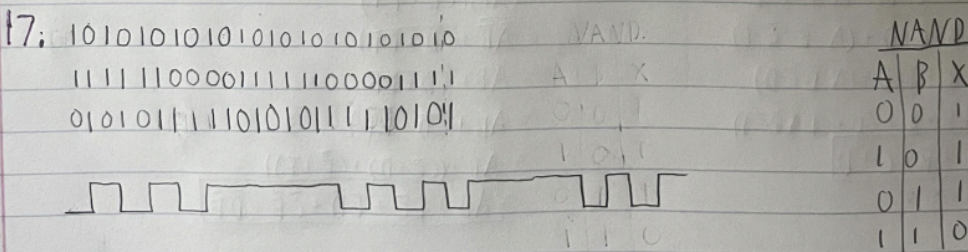
A: 011101110
B: 001000100
X: 001000100

W 6. 

A: 110011001100110
B: 011001100110011
X: 101110111011101
AND: 010001000100010

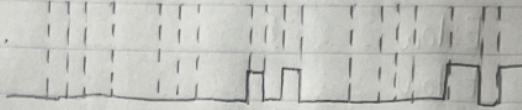
W 11. 

A: 1110110110
B: 1110110110
X: 1110110110

W 17. 

A: 101010101010101010
B: 11111000011110000111
X: 0101011110101011110101
AND: 11111000011110000111
NAND: 001010101010101010

W 22.



NOR

A	B	C	X
0	0	0	1

W 26. A: 0101010101010101

B: 1111110001111111

X: 101010110110101010

XOR

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

10.

a) $\overline{A}\overline{B}(\overline{C}+\overline{D})$

$= \overline{A} + \overline{B} + \overline{C} + \overline{D}$

$= \overline{A} + \overline{B} + \overline{C} + \overline{D}$

$= \overline{A} + \overline{B} + (\overline{C} \cdot \overline{D})$

$= \overline{A} + \overline{B} + (\overline{C} \cdot \overline{D})$

b) $\overline{A}\overline{B}(\overline{C}D + \overline{E}F)$

$= \overline{A} + \overline{B} + (\overline{C}D + \overline{E}F)$

$= \overline{A} + \overline{B} + (\overline{C} \cdot \overline{D} + \overline{E} \cdot \overline{F})$

$= \overline{A} + \overline{B} + ((\overline{C} + \overline{D}) \cdot (\overline{E} + \overline{F}))$

$= \overline{A} + \overline{B} + (\overline{C} + \overline{D}) \cdot (\overline{E} + \overline{F})$

c) $\overline{(A + \overline{B} + C + \overline{D})} + \overline{A}\overline{B}\overline{C}\overline{D}$

$= (\overline{A} \cdot \overline{\overline{B}} \cdot \overline{C} \cdot \overline{\overline{D}}) + (\overline{A} + \overline{B} + \overline{C} + \overline{\overline{D}})$

$= (\overline{A} \cdot B \cdot \overline{C} \cdot D) + (\overline{A} + \overline{B} + \overline{C} + D)$

$= \overline{A} + (\overline{A} \cdot B \cdot \overline{C} \cdot D) + \overline{B} + \overline{C} + D$

$= \overline{A} + \overline{B} + \overline{C} + D$

d) $\overline{(\overline{A} + B + C + D)}(\overline{A}\overline{B}\overline{C}\overline{D})$

$= \overline{\overline{A} + B + C + D} + \overline{A}\overline{B}\overline{C}\overline{D}$

$= (\overline{A} + B + C + D) + \overline{A} \cdot \overline{B} \cdot \overline{C} \cdot \overline{D}$

$= \overline{A} + B + C + D + (\overline{A} \cdot \overline{B} \cdot \overline{C} \cdot \overline{D})$

$= \overline{A} + B + C + D + (\overline{A} \cdot \overline{B} \cdot \overline{C} \cdot \overline{D})$

$= \overline{A} + B + C + (D + (\overline{A} \cdot \overline{B} \cdot \overline{C}))$

$= \overline{A} + B + C + D$

e) $\overline{A}\overline{B}(\overline{C}D + \overline{E}F)(\overline{A}\overline{B} + \overline{C}\overline{D})$

$= (\overline{A}\overline{B}(\overline{A}\overline{B} + \overline{C}\overline{D}))(\overline{C}D + \overline{E}F)$

$= (\overline{A}\overline{B})(\overline{C}D + \overline{E}F)$

$= \overline{A}\overline{B} + (\overline{C}D + \overline{E}F)$

$= \overline{A}\overline{B} + \overline{C}D + \overline{E}F$

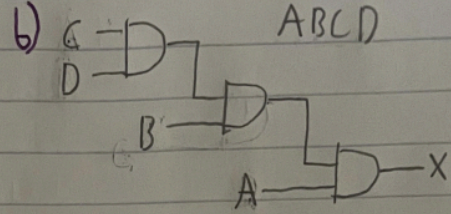
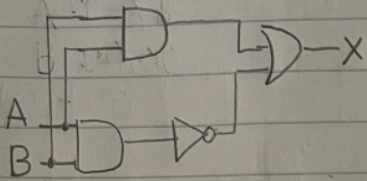
$= \overline{A}\overline{B} + (\overline{C}D \cdot \overline{E}F)$

$= \overline{A}\overline{B} + ((\overline{C} + \overline{D}) \cdot (\overline{E} + \overline{F}))$

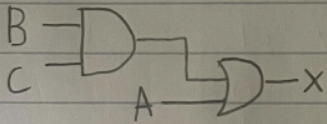
$= \overline{A}\overline{B} + \overline{E}\overline{C} + \overline{E}\overline{D} + \overline{C}\overline{F} + \overline{D}\overline{F}$

W 15.

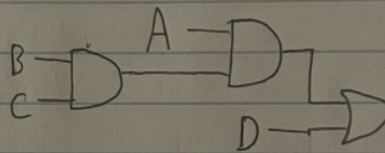
a) $AB + \bar{A}\bar{B} = AB + \bar{A} + \bar{B} = B + \bar{A} + \bar{B}$



c) $A + BC \rightarrow A \text{ or } (B \text{ and } C)$

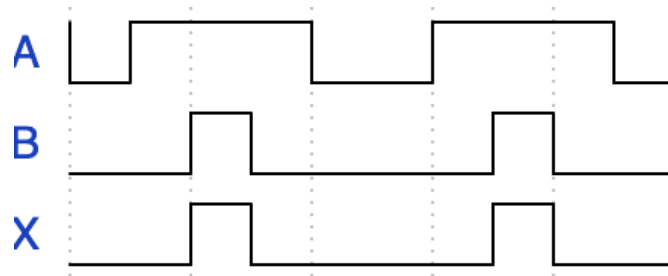


d) $ABC + D \rightarrow (A \text{ and } B \text{ and } C) \text{ or } D$



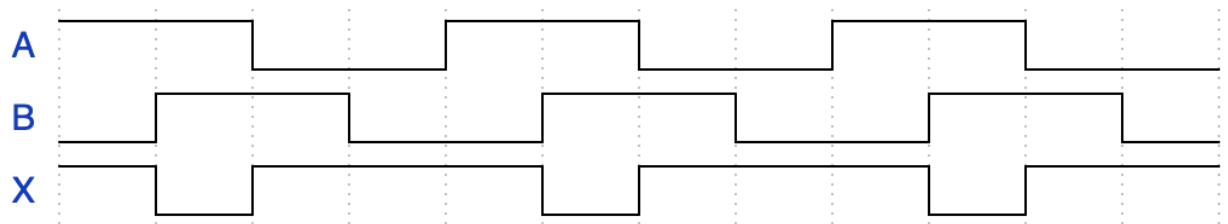
5.

```
{ signal: [
  { name: "A", wave: 'nhlhp' },
  { name: "B", wave: 'lpnl' },
  { name: "X", wave: 'lpnl' }
]}
```



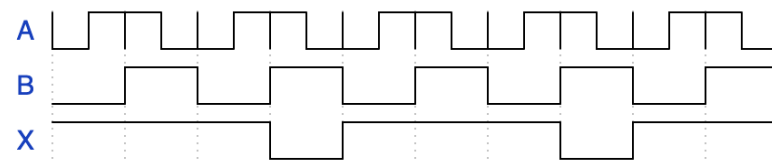
6.

```
{ signal: [
  { name: "A", wave: 'hlhlhlhl' },
  { name: "B", wave: 'nnpnpnpnp' },
  { name: "X", wave: 'phphphph' }
]}
```



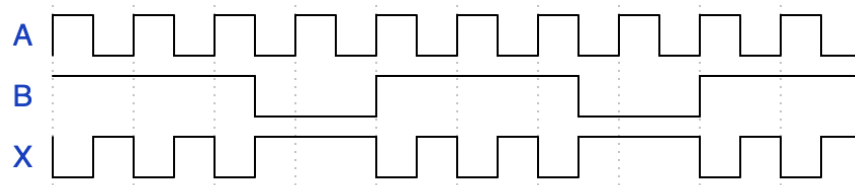
11.

```
{ signal: [
  { name: "A", wave: 'hlhlhlhlhl' },
  { name: "B", wave: 'nnpnpnpnp' },
  { name: "X", wave: 'hphphphph' }
]}
```



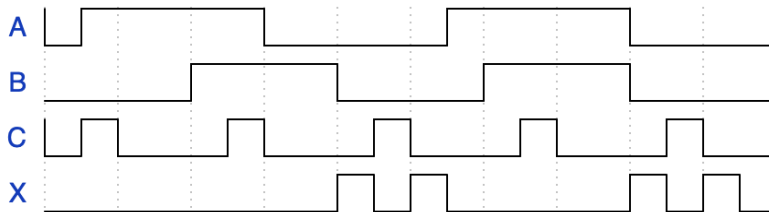
17.

```
{ signal: [
  { name: "A", wave: 'pppppppppp' },
  { name: "B", wave: 'h.plh.plh.' },
  { name: "X", wave: 'nnnhnnnhnn' }
]}
```



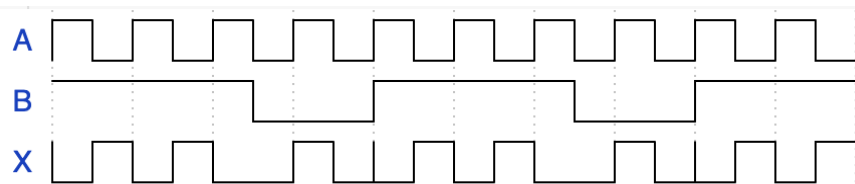
22.

```
{ signal: [
  { name: "A", wave: 'nh.l.nh.l.' },
  { name: "B", wave: 'l.h.l.h.l.' },
  { name: "C", wave: 'nlnlnlnl' },
  { name: "X", wave: 'l...ppl.p.' }
]}
```



26.

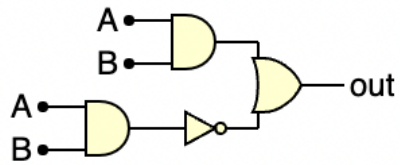
```
{ signal: [
  { name: "A", wave: 'pppppppppp' },
  { name: "B", wave: 'h.plh.plh.' },
  { name: "X", wave: 'nnnhnnnhnn' }
]}
```



15.

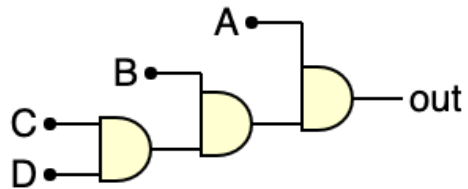
A)

```
{ assign:[  
  "out",  
  ["|", ["&", "A", "B"], ["~", ["&", "A", "B"]]  
]  
]  
}]
```



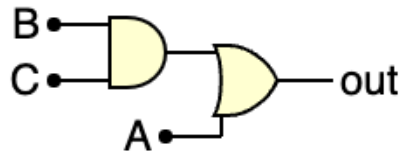
B)

```
{ assign:[  
  "out",  
  ["&", "A",  
    ["&", "B",  
      ["&", "C", "D"]  
    ]  
  ]  
]  
]  
}]
```



C)

```
{ assign:[
  ["out",
    ["|",["&", "B","C"],"A"
  ]
]
}]}
```



D)

```
{ assign:[
  ["out",
    ["|", "D",
      ["&", "A",
        ["&", "B", "C"
      ]
    ]
  ]
]
}]}
```

